

Raphaël Azorin

Senior Machine Learning Engineer | Research to Business ROI

7+ years of industry experience building production pipelines for millions of users.
Hybrid profile combining a Machine Learning PhD with an MSc in Management.

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EXPERIENCE

Ubisoft — *Senior Machine Learning Engineer (full-time)* 2024 - Present

- Developed end-to-end ML systems for fraud detection, including detecting 10K threat actors across millions of nodes, real-time toxicity detection for live chats, and payment fraud models processing thousands of daily transactions.
- Partnered with a team of ML, data, and software engineers to design, deploy, and monitor pipelines in production.
- Tech stack: Python, PyTorch, Spark, Scikit-learn, MLflow, Databricks, MongoDB, Neo4j

Huawei — *Research Engineer (full-time)* 2021 - 2024

- Owned industrial PhD research projects within a dedicated team for network measurements, 8 publications in top venues.
- Developed novel algorithms across probabilistic data structures, tabular time-series, multi-task learning, and resource-constrained machine learning for networked systems. State-of-the-art with up to 11x memory footprint reduction.
- Tech stack: Python, PyTorch, Pandas, Numpy, Transformers, ONNX

xCo Analytics — *Data Engineer (work-study)* 2019 - 2020

- Collaborated with data scientists and engineers to build production ETL handling terabytes of ticketing & IoT telemetry.
- Ensured data quality and infrastructure to support downstream analytics and ML initiatives. Cut model update delay by 4x.
- Tech stack: Python, Scikit-learn, MLflow, Hive, Cassandra, Superset, Databricks, Azure

Marionnaud — *Data Miner (work-study)* 2018 - 2019

- Worked alongside data analysts at this major retailer to build customer behavior models, segmentations, and forecasting.
- Developed clustering and predictive models that directly supported a major product launch targeted at 500K customers.
- Tech stack: Python, Scikit-learn, Oracle SQL Developer

Markentive — *Automation Consultant (full-time)* 2016 - 2018

- Managed client-facing technical projects with C-level executives, directors, and teams of consultants, web devs, and designers.
- Helped scale France's first HubSpot Diamond Partner from 5 to 30 employees by designing automated marketing workflows.
- Tech stack: Python, HubSpot, Marketo, WordPress, Google Analytics

SELECTED PROJECTS & PUBLICATIONS

Dumbo ([GitHub](#)) Tiny classifier inside network devices to predict flow length and improve measurement tasks (CoNEXT'24).

Fieldy ([GitHub](#)) Fine-grained attention in hierarchical transformers for tabular time-series (KDD'24 Workshop).

SchopGPT ([GitHub](#)) Argumentative LLM agent fine-tuned on Schopenhauer's dialectic (HuggingFace & Mistral hackathon).

EDUCATION

PhD in Machine Learning — *Sorbonne University* 2024

MSc in Machine Learning — *PSL University (salutatorian)* 2020

BSc in Computer Science — *PSL University (valedictorian)* 2018

MSc in Management — *Grenoble School of Business / University of Mannheim* 2015

SKILLS & INTERESTS

- French (native) | English (fluent) | Spanish (intermediate)
 - GNNs | LLMs | Transformers | Time-series | Multi-task learning | Fraud detection
- Bouldering (6 y.), boxing (4 y.)
 - Wikipedia contributor